Exercise 2

Solution

1. COUNT() Function

```
SELECT COUNT(*) AS total_number_of_employees
FROM EMPLOYEE
```

Output

total_number_of_employees
10

2. SUM() Function

```
SELECT SUM(salary) AS total_salary
FROM EMPLOYEE
WHERE department = 'IT'
```

Output

total_salary
220000

3. AVG() Function

```
SELECT AVG(salary) AS avg_salary
FROM EMPLOYEE
WHERE department = 'HR'
```

Output

avg_salary
49500

4. MIN() and MAX() functions

```
SELECT MIN(Salary) as MIN_Salary,
       MAX(Salary) as MAX_Salary
FROM EMPLOYEE
```

Output

MIN_salary	MAX_Salary
48000	62000

5. GROUP BY Statement

```
SELECT department
       SUM(Salary) AS total_salary
FROM EMPLOYEE
GROUP BY department
```

Output

department	total_Salary
IT	220000
HR	99000
Finance	119000
Marketing	105000

6. GROUP BY and COUNT()

```
SELECT city,
       COUNT(*) as total_number_of_employees
FROM Employee
GROUP BY city
```

Output

City	total_number_of_employee
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

```
SELECT department
       avg(salary) as avg_salary
FROM EMPLOYEE
Group by department
Order by avg_salary desc;
```

Output

department	average_salary
Finance	59500
IT	55000
Marketing	52500
HR	49500

8. Having Clause

```
SELECT department,
       sum(salary) AS total_salary
FROM EMPLOYEE
GROUP BY department
HAVING sum(salary) > 100000;
```

department	total_salary
IT	220000
Finance	119000
Marketing	105000

9. Combining Group By, Having and ORDER BY

```
SELECT city,
        COUNT(*) AS number_of_employees
FROM EMPLOYEE
Group by city
HAVING COUNT(*) > 1
ORDER BY number_of_employee DESC;
```

~~Output~~

~~| department |~~
~~| --- |~~
~~| IT |~~
~~| Finance |~~
~~| Marketing |~~

Output

City	number_of_employee
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10.
```
SELECT department,
        AVG(salary) AS avg_salary
FROM EMPLOYEE
Group by department
ORDER BY ave_salary DESC
LIMIT 1;
```

Output

department	avg_salary
Finance	59500